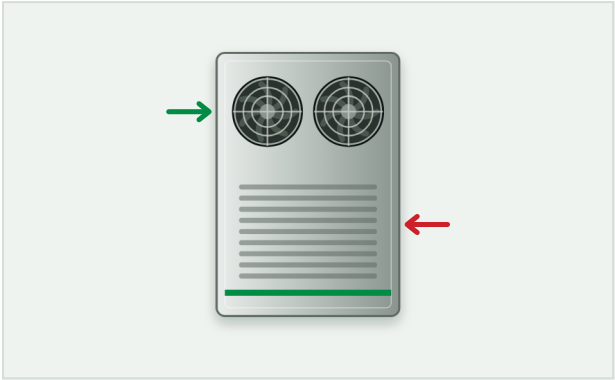
BOSARI
THERMAL MANAGEMENT

DATA SHEET

COLUMBUS 1000

Heat exchanger for electronic cabinets



- **100 W/K cooling efficiency**
- **IP55 and NEMA 12**
- **Fans 115 V/60 Hz, 220 V/50 Hz or DC 12-24-48 V**

TECHNICAL CHARACTERISTICS	
Cooling efficiency	**100 W/K**
Dimensions	**440 × 1050 × 106 mm**
External case material	**Al 99.5%**
Exchanger core material	**Pure aluminium**
Operating temp. range	**−40 °C ÷ +100 °C**
Weight	**19 kg approx.**
Inlet fans	**2 × Ø 172 mm**
Outlet fans	**2 × Ø 172 mm**
Typ. air flow speed	**3.5 m/s**
Typ. noise @3.5 m/s	**58 dB(A)**
EMC compliances	**EN55022 · EN50082-2 · ETS300 342-2**

Description

The COLUMBUS family of passive heat exchangers has been specifically designed for electronic cabinets and environmental chambers. It features an innovative, extremely lightweight construction system, and an easy-to-mount chassis.

These highly efficient units, developed upon the concept of counterflow heat exchangers, have two separated hydraulic circuits, with no air exchange between the cabinet and the ambient.

They comply with IP55 and NEMA 12 requirements.

They can be mounted in any position, offering a great flexibility in designing devices, and they are maintenance free.

The COLUMBUS heat exchangers can be equipped with standard 115V/60Hz or 220V/50Hz fans, or with DC brushless fans (12-24-48V), all provided with finger guards.

The modules feature a thermal switch set at 45°C ± 5°C and NTC sensor for external temperature control. Custom-tailored control systems are available upon demand.

All the electrical connections are available on standard 3x2mm faston terminals.

Materials and components used in the manufacturing of Bosari Thermal Management's COLUMBUS heat exchangers are carefully selected and quality controlled, to ensure the compliance with the requested specifications.